

Brainies Book club

A thousand brains by Jeff Hawkins (Session 3)

Presented by Najmeh Nayyer

"Brilliant.... Exhilarating."
—RICHARD DAWKINS, from the foreword

A THOUSAND BRAINS



A NEW THEORY OF
INTELLIGENCE

JEFF HAWKINS

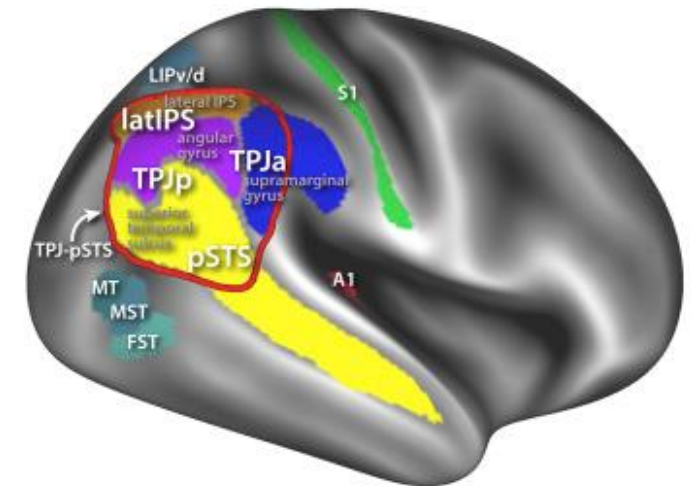
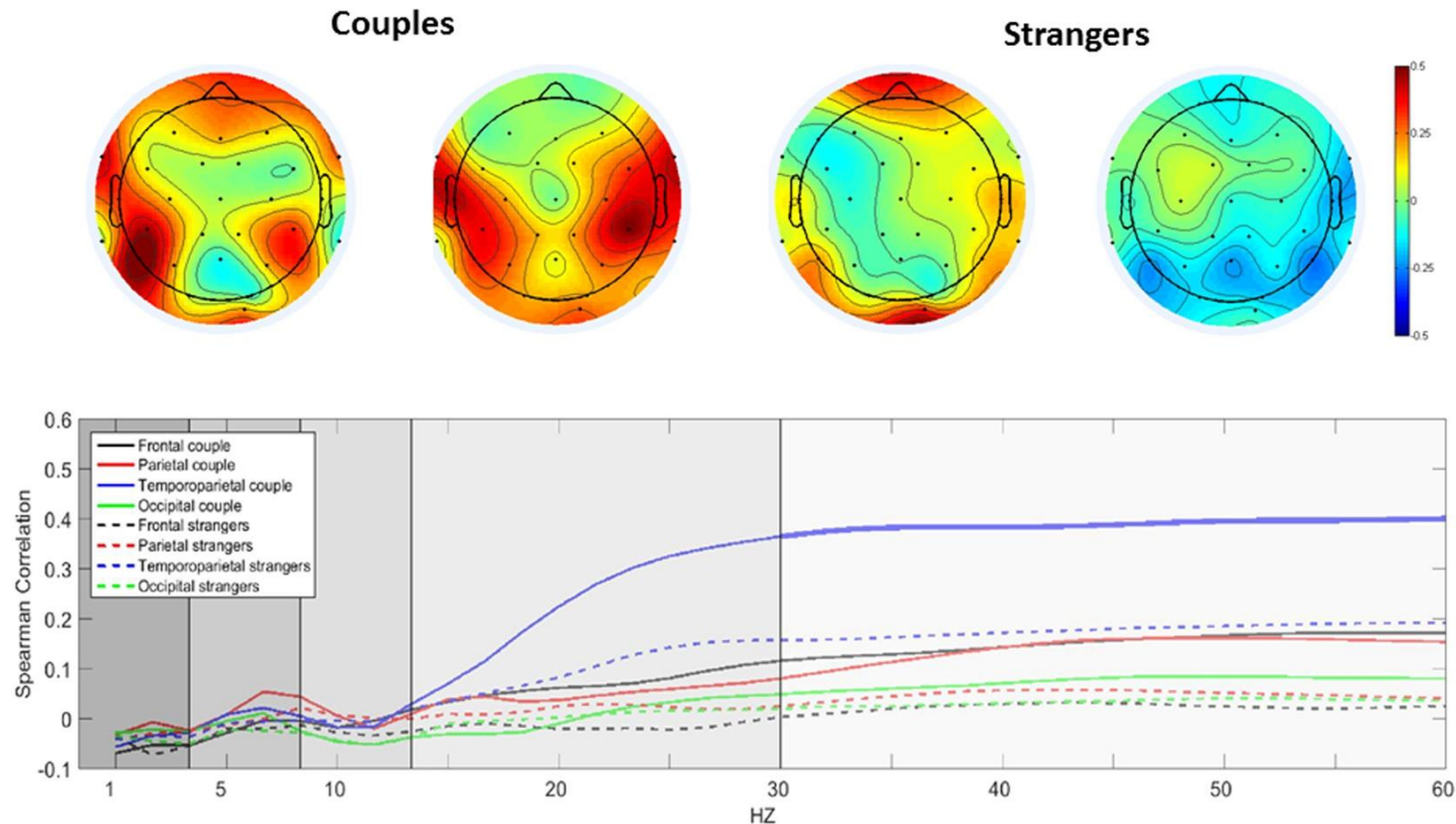
Chapter 3

A Model of the World in Your Head

Your Brain is a prediction machine

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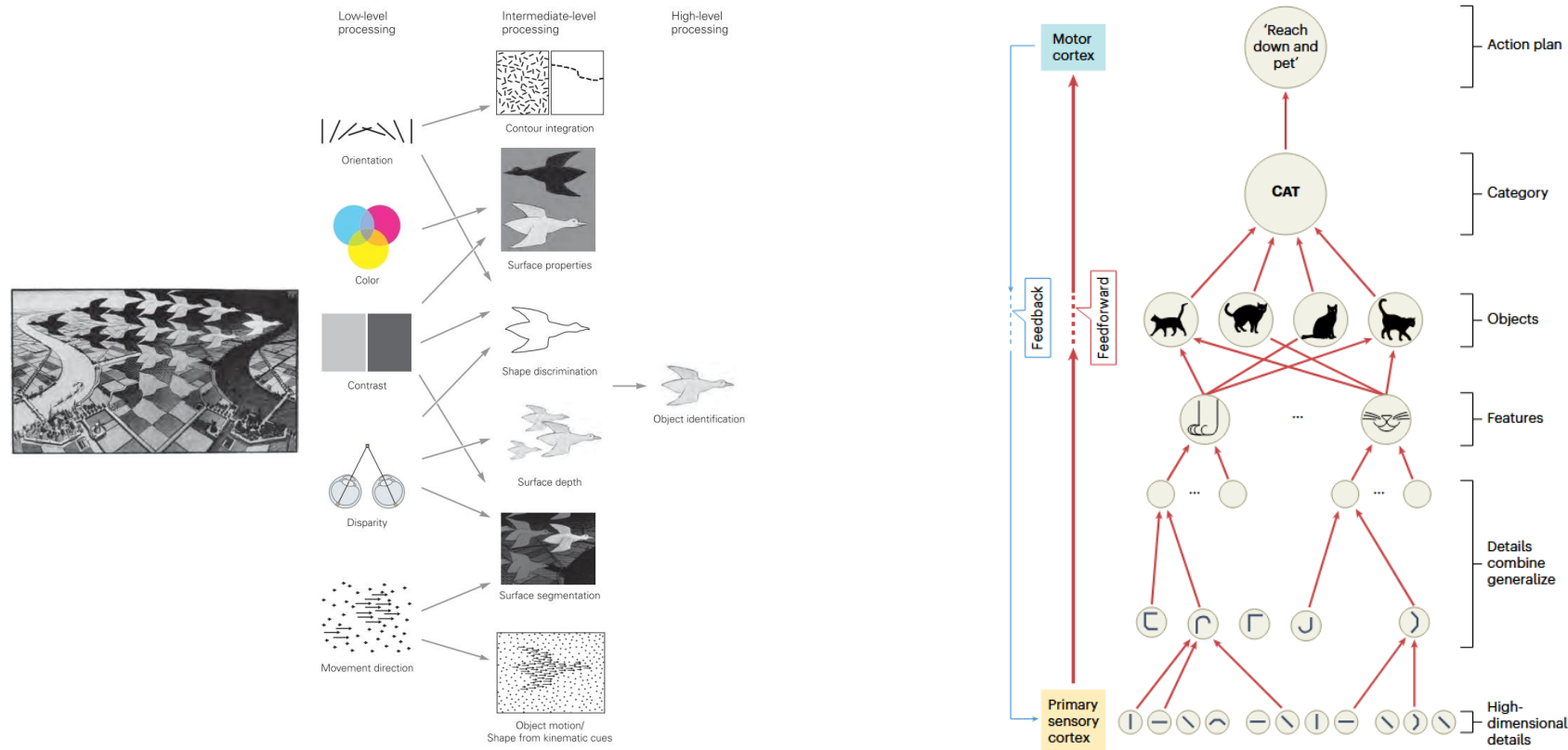
- Noticing change even when we don't look for it.
- The brain wasn't always believed to be a prediction machine
- Telepathy: Interpersonal Neural Synchronization (INS)



Levels of sensory features

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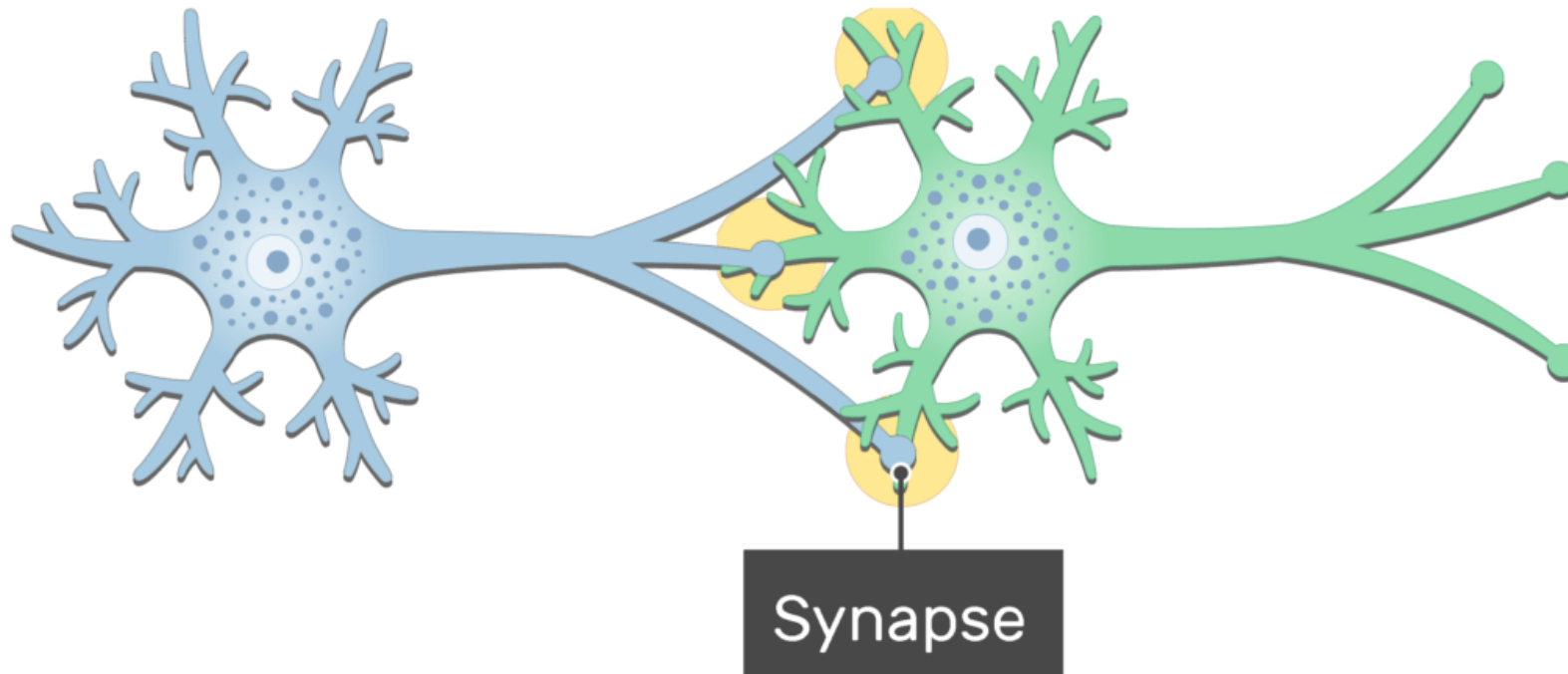
- In each and every level of sensory processing, brain works based on the model and specifies the violations.
- We don't pay attention to an input unless it's violated.
- Both the world and we are continuously changing.



How does the neocortex learn a predictive model of the world?

3

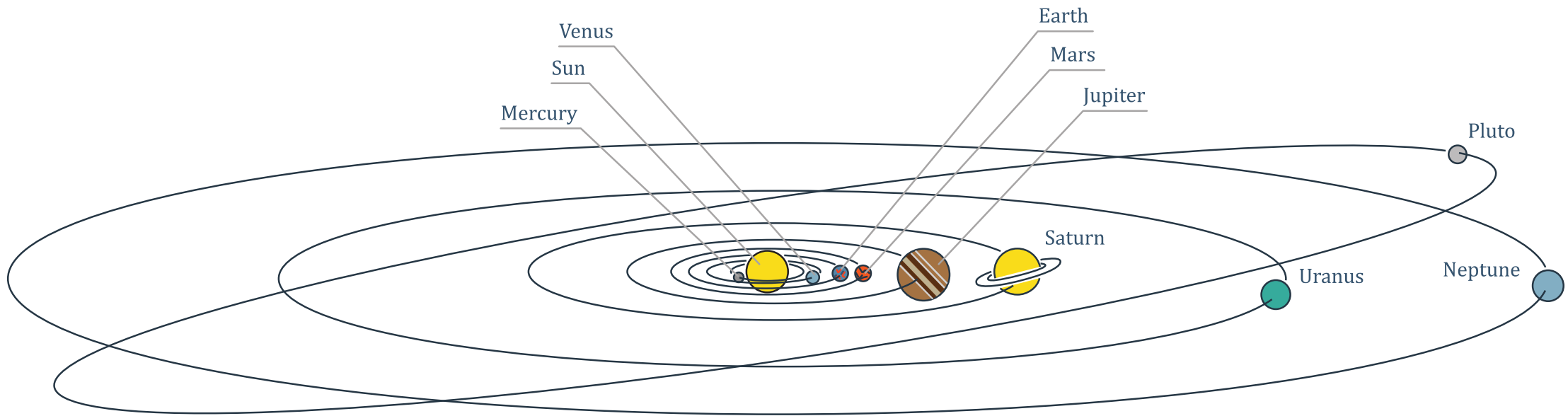
- All of our thoughts and knowledge are stored in neurons, only 2% of which activate to achieve a function.



What is missing?

4

- In the past, people were confused by random and widespread movement of the planets.
- Now, with the right framework (Moving around the sun), even a child can understand it.
- Hawkins discovered three key facts that help us achieve this framework.
 1. The neocortex learns a predictive model of the world
 2. Predictions occur inside neuron
 3. The Secret of the Cortical Column Is Reference Frames



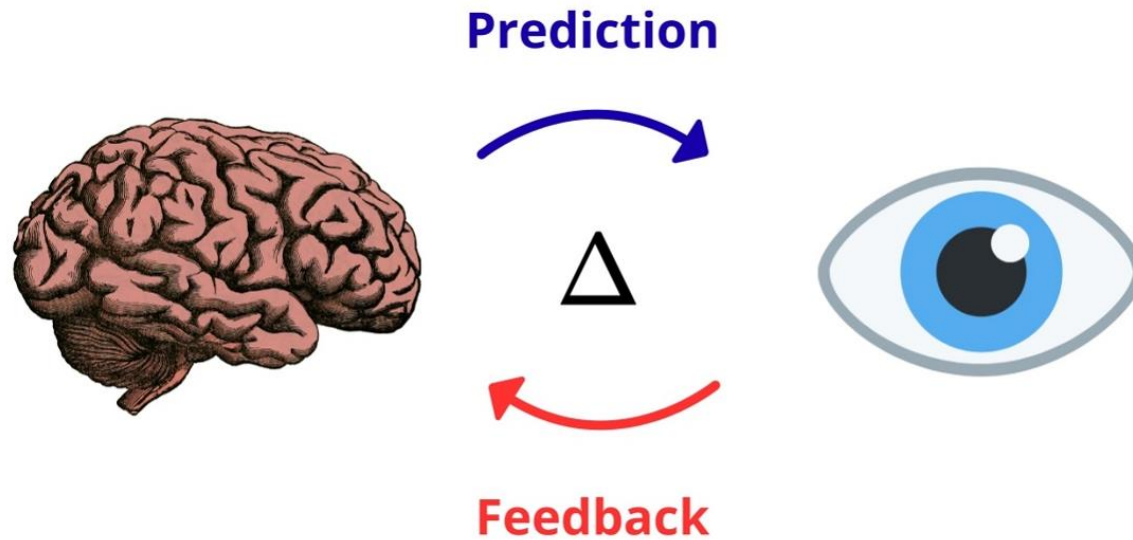
Chapter 4

The Brain Reveals Its Secrets

1. The neocortex learns a predictive model of the world

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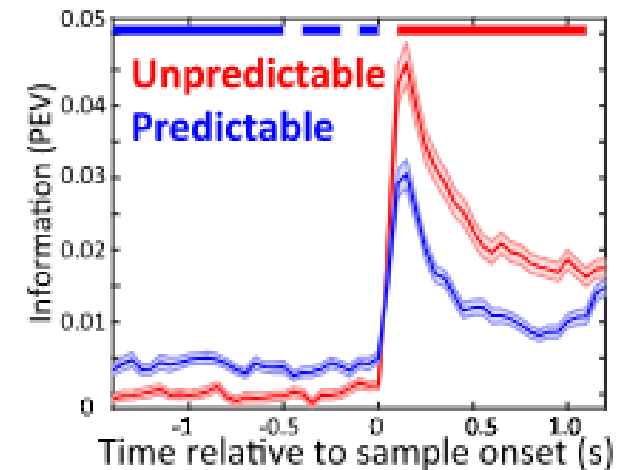
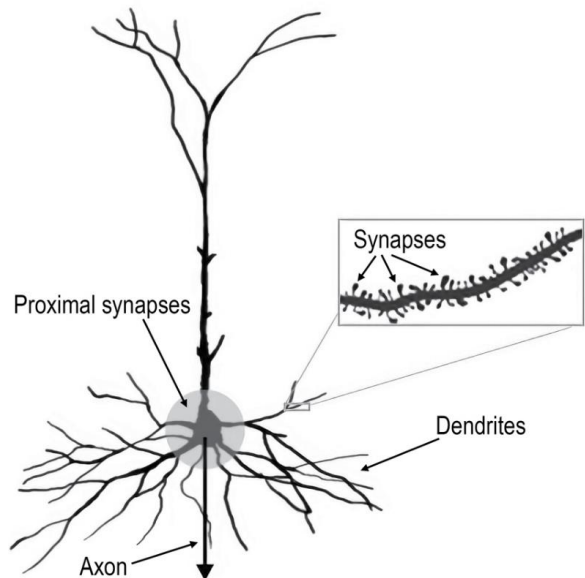
- Possibility: Perception neurons and prediction neurons which also avoid hallucination
- Evidence against it: Haven't observed prediction neurons, we should be aware of some of these predictions



2. Predictions occur inside neuron

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- Mountcastle's cortical algorithm: columns must be making both types of predictions
- Finding the principle of prediction by learning how neurons make sequence memories
- Dendrite spikes are predictions after recognition of a pattern in other neuron(s).
- If the input is predicted, only predictive-state neurons fire.
- Top-down feedback signals act as inhibitory filters that suppress excitatory feedforward signals that match them.
- Ephaptic coupling: electrical fields surrounding neurons influence the degree of excitability of other neurons.
- Dendrites may play an important role in generating these electrical fields.

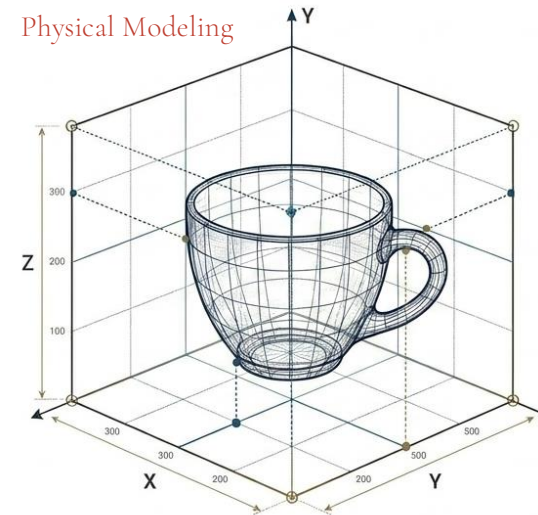


3. The Secret of the Cortical Column Is Reference Frames

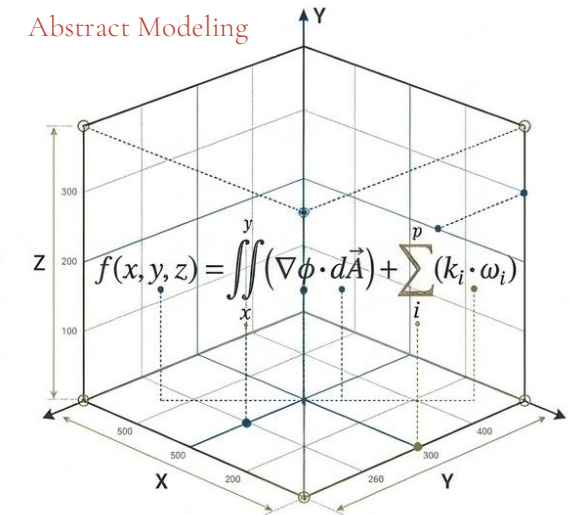
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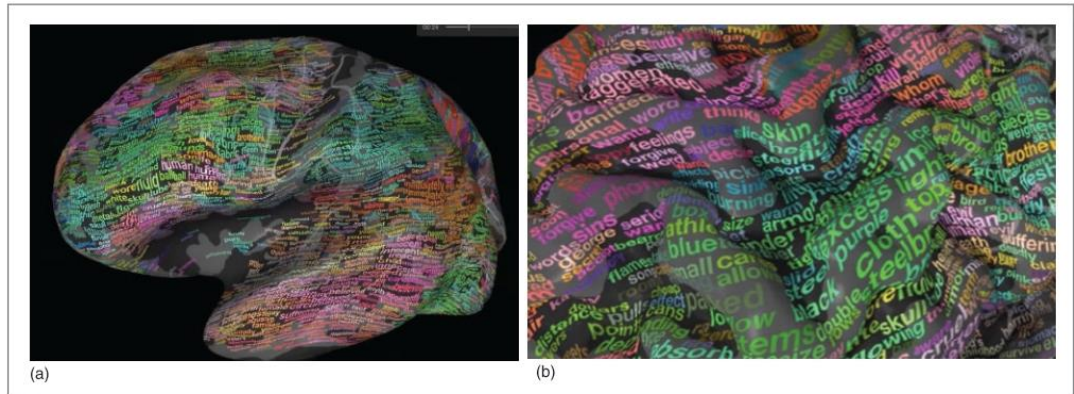
- How does the neocortex predict the upcoming input when we are moving?
 - A robot arm wasn't the answer; a simple cup was!
 - Requirements of prediction: the object, and where your hand is going to be
- Neurons of each sense make their own predictions based on the relative model of the object to the body.
 - Neocortex: reference frame processor rather than sensory processor
- Why are the reference frames so important?
 1. Learning the structure of an object
 2. Manipulating the whole object at once
 3. Planning and creating movement

Physical Modeling

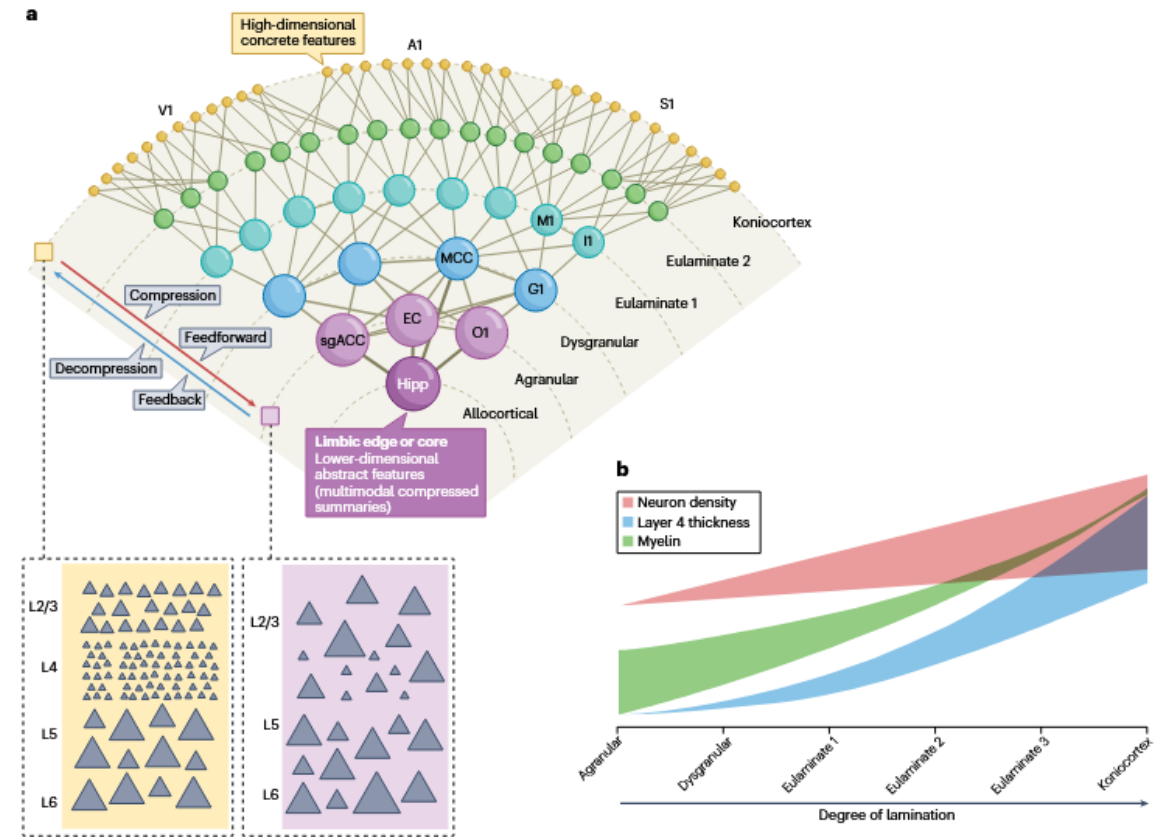


Abstract Modeling





► **Figure 9.24** Results of the Huth et al. (2016) experiment in which participants listened to stories in a scanner. (a) Words that activated different places on the cortex. (b) Close-up of a smaller area of cortex. Note that a particular area usually responded to a number of different words, as indicated in Figure 9.25.



References

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Thank You for Listening!